

Practice Sheet 1.4: Basics of Python Programming

□ MEDIUM LEVEL — IF-ELSE (Exam-Ready)

1. Check Eligibility for Scholarship

Problem Statement:

Write a Python program to check whether a student is **eligible for a scholarship**.

A student is eligible if:

- Marks ≥ 75 **and**
- Family income $\leq 2,50,000$

Input Format:

Marks (integer)

Family income (integer)

Output Format:

Print Eligible for Scholarship or Not Eligible.

2. Electricity Bill Using Slab Rates

Problem Statement:

Write a Python program to calculate the **electricity bill** based on the following slab rates:

- Units $\leq 100 \rightarrow ₹2$ per unit
- Units $\leq 200 \rightarrow ₹3$ per unit
- Units $> 200 \rightarrow ₹5$ per unit

Input Format:

Units consumed (integer)

Output Format:

Print the total bill amount.

3. Check Triangle Validity

Problem Statement:

Write a Python program to check whether three given sides can form a **valid triangle**.

Input Format:

Three integers representing the sides of the triangle.

Output Format:

Print Valid Triangle or Invalid Triangle.

4. Character Classification

Problem Statement:

Write a Python program to check whether the given character is:

- an **uppercase letter**
- a **lowercase letter**
- a **digit**
- a **special character**

Input Format:

A single character.

Output Format:

Print the character type.

5. Calculate Bonus for an Employee

Problem Statement:

Write a Python program to calculate the **bonus** for an employee based on years of service:

- Service ≥ 10 years $\rightarrow$ 10% of salary
- Service ≥ 5 years $\rightarrow$ 7% of salary
- Else $\rightarrow$ 5% of salary

Input Format:

Salary (float)

Years of service (integer)

Output Format:

Print the bonus amount.

6. ATM Withdrawal Validation

Problem Statement:

Write a Python program to check whether a withdrawal is **successful or declined** based on the following conditions:

- Withdrawal amount must be a multiple of 100
- Withdrawal amount must be less than or equal to account balance

Input Format:

Account balance (integer)

Withdrawal amount (integer)

Output Format:

Print Transaction Successful or Transaction Failed.

7. Determine Exam Result with Grace Marks

Problem Statement:

Write a Python program to determine the **exam result**:

- Pass if marks ≥ 40
- Pass with Grace if marks ≥ 35 and < 40
- Fail otherwise

Input Format:

Marks (integer)

Output Format:

Print the result.

8. Find Middle Value of Three Numbers

Problem Statement:

Write a Python program to find the **middle value** among three numbers using if-else.

Input Format:

Three integers.

Output Format:

Print the middle value.

9. Insurance Premium Category

Problem Statement:

Write a Python program to determine the **insurance premium category** based on age:

- Age $< 25 \rightarrow$ High Premium
- Age between 25 and 50 $\rightarrow$ Medium Premium
- Age $> 50 \rightarrow$ Low Premium

Input Format:

Age (integer)

Output Format:

Print the premium category.

10. Check Login Credentials

Problem Statement:

Write a Python program to validate **login credentials**.

Login is successful only if:

- Username is "admin"
- Password is "python123"

Input Format:

Username (string)

Password (string)

Output Format:

Print Login Successful or Invalid Credentials.